

The very first glance on RT/HRT, FLM, and EW

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1 RT/HRT

The idea is, the HEE is equal to the Area, which reflects the duality of CFT(HEE) and AdS(Area).

In an AdS_3/CFT_2 example, take a cylinder as the AdS spacetime bulk, and take a slice over the time axis, get an e^2 , and call e^2 as Σ . Take a piece of Σ , call it A , the rest is A^c as a complementary.

Then, $S(\rho_A) = -Tr(\rho_A \ln \rho_A)$ while $\rho_A = Tr_{A^c}(\rho_{tot})$

RT gives $S(\rho_A) = \frac{Area(\gamma)}{4G\hbar}$.

γ is a surface:

- 1, 2D lower than AdS_n , which is 1D here in AdS_3
- 2, spacelike, each 2 points on it is spacelike.
- 3, minimal surface, average curvature=0
- 4, $\dot{\gamma} = \dot{A}$
- 5, A , γ homology.
- 6, for multiple surfaces satisfy conditions above, pick the one with the minimum area.

Question: Although the minimum area is a certain thing which will not influence the result of S , does the area itself have some special physical meanings? If so, when there are multiple surfaces with the same area, how to pick?

2 FLM

$S(\rho_A) = \frac{Area(\gamma)}{4G\hbar} + S_{field}^{R_A}$. R_A is the area whose boundary is $\gamma + A$.

Also take γ to be the one follows the rules above.

This is the first order correction.

Question: Why don't consider the fields inside bulk but outside R_A ? More generally, why we only focus on the slice instead of the whole bulk?

3 EW

EW changed the standard to choose γ . In the above 6th rule, pick the one which makes $\frac{Area(\gamma)}{4G\hbar} + S_{field}^{R_A}$ a global minimum.

This γ is called Quantum Extremal Surface (QES).

$S_{gen} = \frac{Area(\gamma)}{4G\hbar} + S_{field}^{R_A}$ is called the Generalized Entropy of R_A .

Although FLM is a first order correction, EW is precise in any order. Which is confusing to me at this moment.